

SIROP: High-Level Synthesis with Parallel Patterns

Louis Hildebrand Maheyer Shroff
Christophe Dubach Brigitte Vachon

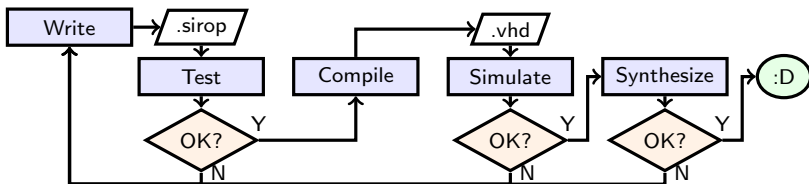


McGill

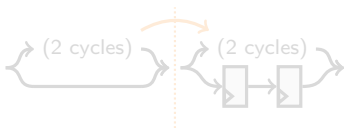
September 22, 2026

Recap: Project Goal

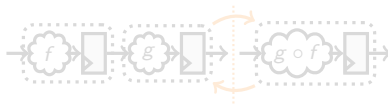
Goal: use a high-level language to develop resource-efficient VHDL implementations of energy reconstruction algorithms



Latency matching



Fusion

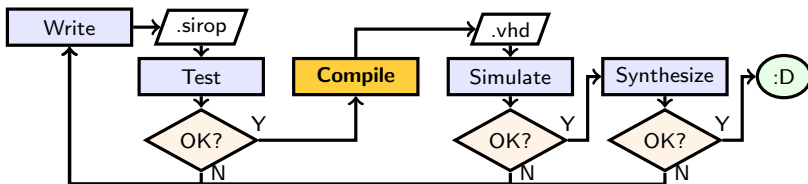


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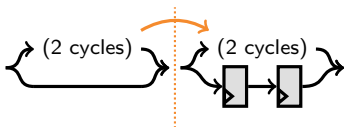
etc.

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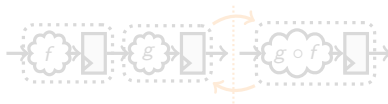


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Fusion

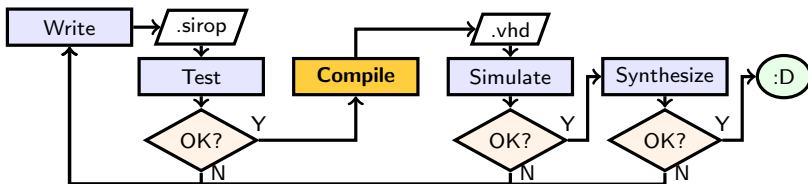
etc.



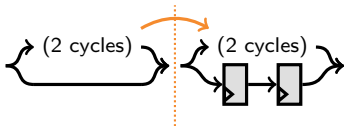
Fission

Recap: Project Goal

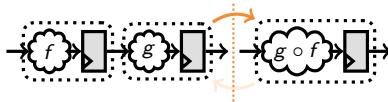
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Latency matching



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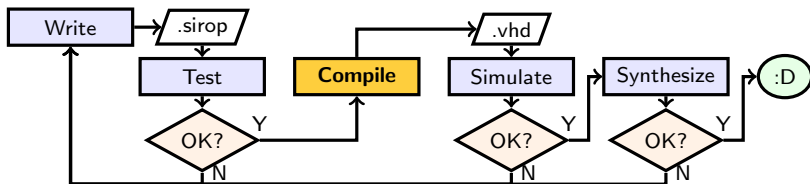


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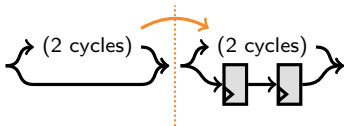
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Recap: Project Goal

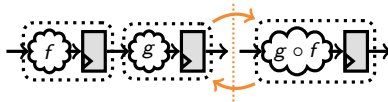
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Fusion



Fission

etc.

Previous Results, dacore FIR Filter

6-tap filter with hard-coded weights, targeting Agilex 7

Design	$f_{\max}$	ALMs	Δ ALMs	DSPs
Handwritten VHDL	880 MHz	123		3
Generated w/ SIROP	823 MHz	118	(-4%)	3

Why the Difference in ALM Usage?

Different DSP pipelining settings

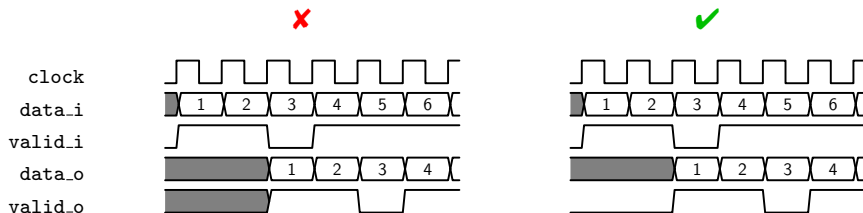
- See presentation explaining DSP register packing:
<https://indico.cern.ch/event/1700332/>
- Ported to LASP-dacore in merge request !143

Updated Results, dacore FIR Filter

6-tap filter with hard-coded weights, targeting Agilex 7

Design	$f_{\max}$	ALMs	Δ ALMs	DSPs
Handwritten, before !143	880 MHz	123		3
Generated w/ SIROP	823 MHz	118	(-4%)	3
Handwritten, after !143	823 MHz	117	(-5%)	3

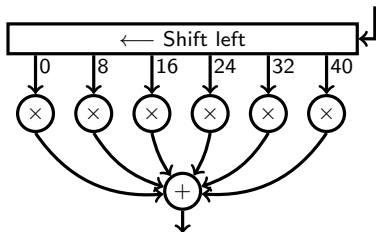
Initial valid Outputs



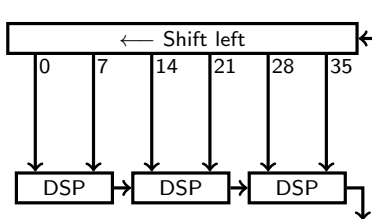
- It's good to ensure the `valid` field in `t_avst_stream` outputs is 0, even immediately after reset
 - See merge request [!142](#) in LASP-dacore
- SIROP can check such conditions (e.g., `valid` bit must be 0 for the first n cycles, where n is the entity's latency)

DSP Input Cascade

Intuitive FIR filter:



Using DSPs:



SIROP can automatically transform the simpler version into the resource-efficient version (by merging and shrinking redundant shift registers)!

Conclusion

- Investigating differences between generated and handwritten VHDL led to ideas for improving resource-efficiency (!143) and robustness (!142)
- SIROP generates a resource-efficient VHDL implementation (using the Agilx DSPs) from concise source code
- Next steps: more advanced energy reconstruction algorithms
 - Neural networks (CPPM, Dresden)
 - Optimal filter with forward corrections (McGill)
- Future direction: faster exploration of neural net architectures (e.g., representation of weights, hypernetworks)



(SIROP
repository)